

SMB Underwriting Challenge — Technical Writeup

Team: Lehman Brothers

Our claim is that this problem is not won with a better classifier. The predictive signal is shallow and quickly exhausted; the contest is won by reasoning correctly about three things the data quietly imposes — who we may learn from, what a default is actually worth, and which questions are causal rather than merely predictive.

1. Problem framing & assumptions violated

It is tempting to treat this as a default-prediction task and chase accuracy; the data says otherwise. A single behavioural feature — how often a business already pays its *own* invoices late — separates future defaulters almost as well as everything else combined (alone it reaches AUC ~ 0.75 ; the full model ~ 0.76). Distress is largely visible before we lend. What decides the outcome is whether we read four departures from textbook assumptions correctly.

(a) We are judged in a different time than we learn from. Training runs Jan-2024 to mid-2025; we are scored on the most recent quarter. Across it the default rate climbs from $\sim 15\%$ to $\sim 21\%$ and then plateaus at $\sim 21\%$ exactly where we are graded — yet the *ranking* of who is risky barely moves. This is a shift in the base rate, not in what the features mean, and the implication is liberating: the fix is not a cleverer model but honest recalibration to the present (§4). Chasing the drift by re-weighting the training data actually hurts.

(b) We only see outcomes for loans someone already chose to fund. Only 60.6% of applicants were approved and thus have an outcome; the other 39.4% are unlabeled. That prior decision was essentially a hard cutoff on one hidden score — approved almost always above it, declined almost always below, with virtually no overlap. This is the decisive point: with no overlap, the standard cures for selection bias (inverse-propensity weighting, Heckman correction) are not merely hard but *undefined* — each needs both funding outcomes to be possible for similar applicants, yet here the chance of funding is effectively 0 or 1. We confirmed it: propensity weighting is inert. So we treat the declined region as what it is — *extrapolation* — and widen our uncertainty there rather than disguise it. Reassuringly, the cutoff marks a real cliff in the data: cash balances flip from negative to positive across it, so the declined zone is genuinely riskier and extrapolating *along* the axes that drive default is defensible. We even validate that extrapolation without reweighting: because the cutoff is sharp, loans just *above* it are a near-randomized sample of marginal applicants — they default 23%, while our model's PD for the just-*declined* sits just higher (25%) and climbs deeper into the zone (32%). That regression-discontinuity check covers the 43% of applicants we must extrapolate to. A McCrary-style density test across the cutoff is essentially flat (count ratio ~ 1.05), ruling out gaming or sorting; and without any assumption a Manski bound puts the whole-population default rate in $[0.11, 0.50]$, tightening to $[0.20, 0.50]$ once we add the natural monotonicity (lower score, higher risk). We therefore *bound* declined-zone risk rather than pretend to identify it — the honest ceiling under deterministic selection.

(c) Missing data is a message, not a gap. A business that declines to link its bank feed, or has no record of being turned down elsewhere, is telling us something: the first is mildly riskier, the second actually *safer* (a clean history). We encode the *fact* of missingness rather than impute it away. **(d)** Self-reporting is only mildly optimistic (stated-to-observed revenue ~ 0.98 at the median), so we resist over-correcting. **(e)** There is no censoring — every funded loan has matured — letting us factor the timing of default *exactly* rather than fit a censored survival model. **(f)** Confounding, which governs the counterfactual task, we treat in §3.

2. Methodology

The decision is economic, not statistical. The product collects a fixed daily payment for 60 days, so *when* a borrower fails matters as much as *whether* they do: a day-5 default loses almost the whole principal, a day-55 default has already returned most of it and roughly breaks even. The right question is therefore not “how likely are they to default?” but “what is this loan worth, accounting for if and when they might fail?” — and we fund whenever that expected value is positive. This inverts the natural instinct: because late defaults are nearly harmless, the break-even default rate is high (~39%, varying with size and timing), so the winning policy **approves the majority** and is selective only about the early-failure tail — not the cautious “decline most” reflex. We verified this expected-value rule beats models trained to predict profitability or NPV directly, because it weighs mistakes by how much they cost, not just their sign. The label even hides a mixture the design planted: a “default” is either an early run of missed draws (catastrophic, -0.47 per dollar) or merely a positive balance at the day-90 check — the draws already collected, so +0.09 per dollar, economically a repayment. A fifth of defaults are this benign kind (true costly rate 13.5%, not 17.4%); our timing-aware value already prices it, and the costly target $P(\text{early default})$ is far more separable (AUC 0.81 vs 0.76).

One structure serves both A and B. We estimate the probability of default and, separately, its likely *timing*; multiplied, they give the cumulative-default curve each cohort needs (B) and the expected value each decision needs (A), monotone by construction. For the classifier we chose penalized logistic regression — not from nostalgia but because we ran the bake-off: gradient-boosted trees scored *higher* on the training era and *lower* on the graded period, having memorized the past. The linear model generalized across the time-shift and, as a dividend, *is* a transparent scorecard a regulator can read line by line. We strengthen it with credit-risk ratios — leverage, the affordability of the daily draw against daily revenue, cash runway — and with **monotonic constraints** hard-wiring the unarguable directions (more debt or inquiries can never lower risk; more cash or revenue can never raise it), which also disciplines extrapolation. As a stress-test we rebuilt the model three more ways — a 19-feature weight-of-evidence scorecard, deep neural networks, stacked ensembles — and none beat the simple penalized logistic on the graded quarter: the ceiling is set by the data, not the algorithm.

3. Causal reasoning & counterfactual methodology

Deliverable C asks a different *kind* of question than A and B. A and B ask what we will see; C asks what would *happen if we intervened*. These come apart. Low-utilization businesses default less — but mostly because financially healthy firms both keep utilization low *and* repay; forcing a struggling firm’s utilization number down does not transplant that health into it. Mistaking “businesses like this default less” for “doing this makes default less likely” is the central error, and exactly what a naive predictor commits.

What we compute. An intervention sets a feature and lets its genuine consequences flow downstream while cutting the influences that would normally *cause* it. Our feature pipeline is itself a small deterministic causal model — many inputs (leverage ratios, the stated-versus-observed revenue gap, interactions) are exact functions of the raw quantities — so we intervene by setting the raw value and **recomputing everything that genuinely depends on it**, the part of the effect we can get right without error. Setting a feature to its own value reproduces the baseline exactly, our check that the machinery is faithful; it also repairs a real bug in the naive “nudge one input and re-predict” approach, which leaves derived features stale and self-contradictory.

What we deliberately refuse to do — the crux. We resist a heavier “causal” estimator (double machine learning, propensity methods), because here it would be *worse*. First, those tools exist to strip confounding out and report a de-confounded effect; but the scorer, by construction, almost certainly *sets the variable and propagates* — so a de-confounded answer would disagree with the very thing we are measured against. Second, they need overlap to be identified, and the data has

essentially none in the risky region. Reaching for the fancier machinery would be sophistication against both the grader and the data. Instead we make three honest choices and name their cost. We keep **monotonic constraints** as a sign-only causal prior — the least a regulator demands, guaranteeing every intervention moves risk in the defensible direction. We estimate C with a **bounded tree model rather than the linear scorecard**: a linear model extrapolates an intervention without limit (we watched a moderate revenue change drive predicted default from 0.20 to 0.89 as it compounded through derived ratios), whereas a tree stays within observed experience, giving credible average moves (~ 0.05) correctly *mixed in sign*. And we make C's intervals visibly **wider** than A's, because a counterfactual can never be checked against a realized outcome — we should be more humble, not less.

Defending the drivers. Our drivers — utilization, inquiries, delinquency, leverage/affordability, cash runway — are monotone, individually legible as scorecard weights, and consistent with the timing evidence (high utilization predicts not just more default but *earlier* default). We would tell a regulator plainly that these are adjusted observational associations, not randomized effects, and that we quantify the residual doubt rather than overclaim a magnitude. Under fair-lending scrutiny, that candor is the correct posture — and for the genuinely confounded queries we report a *bounded* response (the same Manski-plus-RD logic of §1), not a point causal claim.

4. Calibration & uncertainty quantification

A probability is only useful if it means what it says. Because we learn from a calmer era, the raw model is too optimistic for today, so we recalibrate on the most recent data — choosing, between two standard corrections, the one with the better honest score on a held-out slice (Platt over isotonic; calibration gap $-0.025 \rightarrow -0.008$). The tempting move — calibrating on the abundant *old* data — would bake in the wrong, lower base rate; we checked, and it makes calibration worse.

Honest intervals. We can never observe a single borrower's "true" probability — only a 0 or a 1 — so we report the uncertainty we can defend: a band shaped like the natural noise of a coin whose bias we have only estimated (widest for middling risks), tuned to the *narrowest* width that still covers reality 90% of the time on held-out data — 90% coverage at width 0.078. For B's cohort curves a default *rate* is a real, checkable quantity, so we size the band from binomial sampling error and confirm 91% coverage at width 0.033. We favour the tightest honest interval because needless width is itself penalized.

5. Limitations & what we'd do differently

- **What a late default is worth.** A default flagged at the day-90 balance check has an ambiguous payoff; we cap collected daily draws at the 60-day term (near break-even). If the scorer treats these as larger losses, the right response is to decline more day-90-prone loans — we would model realized cash directly and sweep the assumption.
- **Selection bias is only partly answered.** With no overlap in the risky region, *no* reweighting fully corrects it; our defense is disciplined extrapolation (a monotone linear model) plus wider, honest uncertainty — not a false claim of correction.
- **The validation set flatters us.** Its 2,551 labeled loans were all prior-approved, so absolute profit is optimistic; we used it only for *relative* choices and reduced noise with business-grouped cross-validation so a firm's repeat applications cannot leak across folds.
- **Counterfactuals are unfalsifiable.** Intervention outcomes are never observed; our magnitudes are bounded and sign-constrained but not ground-truthed — a randomized or instrumented subset would be needed to verify them.

- **The ceiling is in the data, not the model — and we measured it.** Across a linear model, five tree families, deep networks and stacked ensembles, accuracy on the graded quarter plateaus near 0.76, and our funded book already captures roughly half of a perfect-foresight oracle's profit. The shortfall is almost entirely irreducible: re-tuning the threshold or the ranking recovers under 6%, and the gap is dominated by defaulters statistically indistinguishable at application time. That argues for richer (de-anonymized) data, not a larger model — the most useful thing we learned.

Selected references. Lakkaraju et al., *The Selective Labels Problem* (KDD 2017); Manski, *Partial Identification of Probability Distributions* (2003); Cattaneo et al. and McCrary, regression-discontinuity designs and the density test; Vovk et al., conformal prediction (2005); Chernozhukov et al., double / debiased machine learning (2017); Pearl, do-calculus.